Course outline

About NPTEL

How does an NPTEL online course work?

Week 0

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Week 11

- Finetuning and Jailbreaking: Hands-on
- AI Governance
- Research Overview: SaGE-Quantifying moral consistency in LLMs
- Quiz: Week 11: Assignment 11
- Week 11 Feedback Form: Responsible & Safe AI Systems
- Week 11- Lecture Materials
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Week 11: Assignment 11

The due date for submitting this assignment has passed.

Due on 2024-10-

As per our records you have not submitted this assignment.

1) In the context of the paper SaGE, what is semantic consistency?

- Semantically equivalent questions should yield semantically equivalent answers
- Semantically equivalent questions should yield same answers
- Same questions should yield same answers
- Same questions should yield semantically equivalent answers

No, the answer is incorrect.
Score: 0
Accepted Answers:
Semantically equivalent questions should yield semantically equivalent answers

2) Why do models struggle with tasks in “moral scenarios”?

- Models lack the ability to process large datasets efficiently
- Models prioritise emotion over logic in moral decision-making
- The models are limited by insufficient computational power
- Conflicting training data due to different morals that people have

No, the answer is incorrect.
Score: 0
Accepted Answers:
Conflicting training data due to different morals that people have

3) What metric was used to determine the quality of the paraphrase of the questions in the SaGE paper?

- BERTScore
- Parascore
- Jaccard Similarity
- Cosine Similarity

No, the answer is incorrect.
Score: 0
Accepted Answers:
Parascore

4) How does the SaGE paper relate entropy and consistency?

- More Entropy implies consistency
- Less entropy implies inconsistency
- Less Entropy implies consistency
- More entropy implies inconsistency

No, the answer is incorrect.
Score: 0
Accepted Answers:
Less Entropy implies consistency
More entropy implies inconsistency

5) Identify the statements that are TRUE with respect to the current LLMs.

- LLMs are not consistent in their generation
- A good accuracy on benchmark datasets correlates with high consistency
- LLMs are consistent in their generation
- A good accuracy on benchmark datasets does not correlate with high consistency

No, the answer is incorrect.
Score: 0
Accepted Answers:
LLMs are not consistent in their generation
A good accuracy on benchmark datasets does not correlate with high consistency

6) Why is AI Governance important?

- To prevent AI from learning new tasks independently
- To limit the efficiency of AI in performing complex tasks
- Ensure AI isn't used for unethical acts
- To prevent AI from being used in scientific research

No, the answer is incorrect.
Score: 0

Accepted Answers:
Ensure AI isn't used for unethical acts

7) What are some aspects of AI Governance that are in focus in the current times?

- [ ] Revealing the amount of compute used during training past a certain compute threshold
- [ ] Limiting AI systems to only perform manual labor tasks
- [ ] Ensuring right to erasure
- [ ] Prohibiting the use of AI in any form of automation

No, the answer is incorrect.
Score: 0

Accepted Answers:
Revealing the amount of compute used during training past a certain compute threshold
Ensuring right to erasure

8) Which of the following are key OECD AI Principles?

- [ ] Inclusive growth, sustainable development, and well-being
- [ ] Limiting AI to industrial use cases
- [ ] Transparency and explainability
- [ ] Restricting international AI collaboration

No, the answer is incorrect.
Score: 0

Accepted Answers:
Inclusive growth, sustainable development, and well-being
Transparency and explainability

9) As discussed in the lecture, When you have domain specific task, what kind of finetuning is preferred?

- [ ] Full-model finetuning
- [ ] Layer-specific finetuning
- [ ] Head-level finetuning
- [ ] Retraining

No, the answer is incorrect.
Score: 0

Accepted Answers:
Full-model finetuning
Layer-specific finetuning

10) What are the cons of full-model finetuning?

- ( ) Overfitting
- ( ) Catastrophic forgetting
- ( ) Increase in parameters
- ( ) Change in architecture

No, the answer is incorrect.
Score: 0

Accepted Answers:
Catastrophic forgetting

11) What are adapters?

- ( ) Remove existing layers from a model
- ( ) Convert a model to a simpler architecture
- ( ) Replace the model's original parameters entirely
- ( ) Add additional layers to a preexisting architecture

No, the answer is incorrect.
Score: 0

Accepted Answers:
Add additional layers to a preexisting architecture

12) What is instruction finetuning?

- ( ) Model is trained to ignore user instructions and operate independently based on its previous training
- ( ) Model's training objective is to follow the directions provided by the user when performing the task
- ( ) Process of training a model solely on instruction data without any real-world data
- ( ) Modifying the model's architecture to include specific instructions directly within its layers

No, the answer is incorrect.
Score: 0

Accepted Answers:
Model's training objective is to follow the directions provided by the user when performing the task